13 September 1984
Superseding
MIL-C-81093A(OS)
29 November 1966
and Amendment 5
2 November 1977

MILITARY SPECIFICATION

CARTRIDGES, EXPLOSIVE BOLT

(For Torpedoes Mk 46 All Mods)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments of Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of electrically initiated explosive bolt cartridges.

1.2 Classification. The explosive bolt cartridges shall be of the following types, as specified (see 6.2):

(a) Type I. Explosive Bolt Cartridge Mk 78 Mod 0 (LD 535751, Dwg 2158784).

(b) Type II. Explosive Bolt Cartridge, Dwg 2539121.

1.2.1 When requirements herein refer to the ~'cartridge,"the requirements apply to either the Type I or Type II cartridge. When requirements are pertinent to only one type of cartridge, the appropriate cartridge is designated.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficlal comments (recommendations, additions, deletions) and any pertinent data which may be of use In improving this document should be addressed to: Commanding Officer, Naval Ordnance Station (524), Indian Head, MD, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document,or by letter.

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SPECIFICATIONS

MILITARY

MIL-R-398	RDX

STANDARDS

FEDERAL

FED-STD-H28	Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-202	Test Methods for Electronic and Electronic Component Parts
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS 21083	Nut, Self-Locking, Hexagon, Non-Metallic Insert, Low Height, 250°F
MS 28775	Packing, Preformed, Hydraulic, Plus 275°F (O-ring)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

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DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

LD 535751	Cartridge, Explosive Bolt, Mk 78 Mod 0
2158784	Cartridge, Explosive Bolt, Mk 78 Mod 0
2132278	Bolt, Explosive (Machined)
2539121	Cartridge, Explosive Bolt

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

49 CFR 100-199	Transportation

(Copies of specifications, standards, drawings and publications required by contractors in connections with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.2). The sample shall:

(a) Be manufactured using the procedures proposed for production.

(b) Consist of 50 cartridges.

(c) Be tested as specified in Section 4 herein and is for the purpose of determining, prior to starting production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Basic requirements

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3.2.1 Materials and treatment of materials and parts. Unless otherwise specified in the contract or order, materials and treatments of materials and parts used in the manufacture of the cartridges shall be as specified herein and the applicable documents listed in LD 535751 (Type I) or Dwg 2539121 (Type II).

3.2.1.1 Selection of materials. Specifications and standards for materials, parts and Government certification and approval of processes and equipment, which are not specifically designated herein, and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided herein. AN, JAN, MS Standard parts shall be used wherever they are suitable for the purpose, and shall be Identified on the drawing by part number. Commercial utility parts such as screws, bolts, nuts, and cotter pints may be used, provided they possess suitable properties and are replaceable by the AN, JAN, or MS Standard parts without alteration, and provided the corresponding AN, JAN, or MS part numbers are references on the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements specified herein.

* 3.2.1.2 Metals. Metals shall be of a corrosion-resisting type or shall be adequately protected to resist corrosion during the normal service and storage life of the cartridge. Copper or copper bearing alloys capable of being corroded by hydrazoic acid shall not be used within the sealed volume.

* 3.2.1.3 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other. Dissimilar metals shall be as defined in MIL-STD-889.

* 3.2.1.4 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. Parts and materials shall be fungus-resistant in accordance with requirement 4 of MIL-STD-454.

3.2.1.5 Threads. Unless otherwise specified, all threads shall be in accordance with FED-STD-H28.

3.2.1.6 Interchangeability. Parts or subassemblies having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

3.2.2 Gages. The supplier shall provide whatever gages are necessary and adequate to ensure that the materials and parts of the cartridge will meet dimensions and requirements shown on the applicable documents listed in LD 535751 (Type I) or DWG 2539121 (Type II). When requested by the supplier,

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the procuring activity will furnish existing pertinent drawings of Naval final inspection gages for use only in the designing of the supplier's gages. This action does not relieve the supplier of the responsibility of designing and manfacturing the gages needed for the satisfactory fulfillment of the contract requirements.

3.2.3 Inspection and test equipment and facilities. The supplier shall furnish and maintain all necessary inspection and test equipment and facilities, and shall provide personnel for performing acceptance tests specified herein. Test equipment shall be adequate to permit performance of the required acceptance tests.

3.3 Performance and physical characteristics

3.3.1 Brisance rating. The detonation of the cartridge shall produce an explosive output sufficient to break a frangible bolt conforming to Dwg 2132278 at any temperature from 0 to 140 degrees Fahrenheit (°F).

3.3.2 Ignition time. The elapsed time between closing the ignition circuit and detonation of the cartridge shall not exceed 15 millisecond (ins) at any temperature from 0 to 140°F.

3.3.3 Explosive charges. The measuring and loading of the following explosive ingredients of the cartridge shall be consistent with the requirements of the the applicable documents listed in LD 535751 (Type I) or Dwg 2539121 (Type II).

3.3.3.1 Ignition charge. The cartridge shall contain a quanitity of lead styphnate sufficient to ignite the intermediate charge (3.3.3.2) with the firing current described in 3.3.4.1. The lead styphnate, after being loaded into the cartridge, shall be cured in accordance with Dwg 2158784 (Type I) or Dwg 2539121 (Type II).

* 3.3.3.2 Intermediate charges. The second initiating agent shall consist of polyvinyl alchol precipitated lead azide (pva lead azide) having a moisture content of not more than 1.0 percent.

* 3.3.3.3 Base charge. The base charge of the cartridge shall consist of RDX Type I, or alternate: RDX Type II, conforming to MIL-R-398. The moisture content of the RDX shall not exceed 0.13 percent.

3.3.3.4 Loading and assembly. There shall be no evidence of loose powder, fissures In the compressed charges, or incorrect assembly of the cartridge. The loading and assembly techniques employed shall be such as to ensure the moisture content specified in 3.3.3.2 and 3.3.3.3 has not been exceeded prior to hermetic sealing of the cartridge assembly.

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* 3.3.3.5 Hermetic seal. The cartridge shall be hermetically sealed to prevent leakage in excess of 1×10^{-6} atmospheric cubic centimeters per second (ATM cm3/s), Sealing shall be accomplished by soldering or welding.

3.3.4 Firing current

3.3.4.1 Firing. The cartridge shall ignite and detonate in the ignition time required in 3.3.2 when 0.5 ampere current is supplied to one of the two circuits of the Type I or Type II cartridge at any temperature from 0 to 140°F. During first article, periodic production, and acceptance tests, alternate firing circuits shall be selected to ensure conformance to this specification.

3.3.4.2 Nonfiring. When subjected to a constant direct current (de) of 0.1 ampere (maximum) on each bridge circuit for five minutes (minimum), the cartridge shall not fire; the bridge wires shall not burn out; nor shall the cartridge thereafter fail to meet the requirements of 3.3 and 3.4.

3.3.5 Circuit resistance. The resistance of each circuit shall be 2.0 plus or minus 0.6 ohms at 77°F.

3.3.6 Insulation resistance. With the pins shorted, the insulation resistance between the pins and the housing of the cartridge shall be greater than 2 megohms under an applied potential of 500 vdc.

3.4 Environmental requirements. Unless otherwise specified, the cartridge shall be capable of conforming to the requirements of this specification after being exposed to the environments and conditions specified in this section.

3.4.1 Vibration. The cartridge shall be capable of withstanding the vibration shown in TABLE I along the axes shown in FIGURE 1.

TABLE I

Vibration schedule

Condition (see 6.3.4)	Double amplitude (in.) minimum	Acceleration (g's) minimum	Frequency range (cps)
Nonoperating and ready	0.75	--	5-6
	----	1.3	6-16
	0.10	-.	16-24
	---	3.0	24-40
	0.036	--	40-74
	---	10.0	74-500

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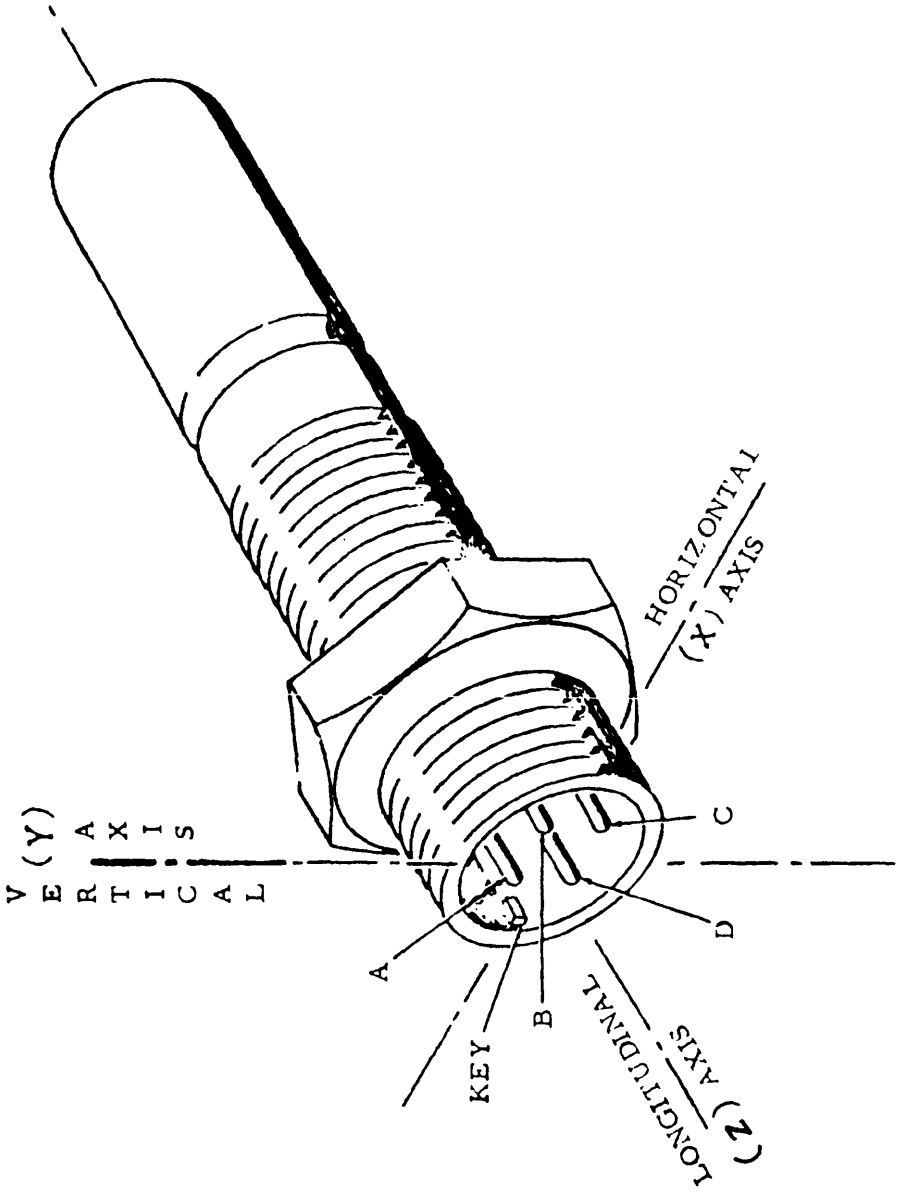


FIGURE 1. Axes of the cartridge.

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3.4.2 Shock (6.3.2). The cartridge shall be capable of withstanding shock loads in accordance with the schedule of TABLE II.

TABLE II

Shock schedule

Condition (see 6.3.3.4)	Peak Magnitude (g's)	Minimum Duration (ins)	Acceleration Direction (Axis) (see FIGURE I)
Nonoperating and ready	60	8.0	Any
	45	20.0	Horizontal and Vertical
	300	0.5	Horizontal and Vertical

3.4.3 Temperature. The cartridge shall be capable of conforming to the requirements of this specification after being at any temperature between minus 45 and plus 140°F in the nonoperating condition (see 6.3.4.1) and while at any temperature between 0 and 140°F in the ready non-operating (see 6.3.4.2) and ready (see 6.3.4.3) condition.

3.4.4 Watertightness. After firing, the insert seal of the cartridge shall be capable of withstanding a hydrostatic pressure of 1000 psi for one minute without leaking.

3.4.5 Humidity. The cartridge shall be capable of withstanding 95 percent relative humidity at 110°F for eight hours.

3.5 Rough handling and safety

* 3.5.1 Jolt test. The cartridge shall be capable of meeting the acceptance criteria of the jolt test of MIL-STD-331, Test Number 101.

* 3.5.2 Jumble test. The cartridge shall be capable of meeting the acceptance criteria of the jumble test of MIL-STD-331, Test Number 102.

* 3.5.3 Drop test. The cartridge shall be capable of meeting the acceptance criteria of the 40 foot drop test of MIL-STD-331, Test Number 103.

3.6 Storage life

3.6.1 Ready. The cartridge, in the ready condition (see 6.3.4), shall be capable of meeting the performance requirements of this specification after one year of unattended storage at temperature from minus 45 to plus 140°F.

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3.6.2 Nonoperating. The cartridge, in the nonoperating condition (see 6.3.4), shall be capable of meeting the performance requirements of this specification after five years of attended storage at temperatures from minus 45 to plus 140°F.

3.7 Identification marking. The cartridge shall be permanently serialized and shall be identified in accordance with MIL-STD-130 and Dwg 2158784 (Type I or Dwg 2539121 (Type II).

3.8 Workmanship. The workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation consistent with the requirements of this specification. Parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contrator is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections l re deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the cartridge shall be classified as follows:

(a) First article inspection

(1) First article sample (see 4.3.1)

(2) First article tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

(c) Acceptance inspection

(1) Acceptance Sampling (See 4.3.3)

(2) Acceptance tests and examinations (see 4.7)

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4,3 Sampling

* 4.3.1 First article sample. A first article sample of 50 cartridges, manufactured in accordance with 3.1, shall be subjected to the first article tests detailed in 4.5 at a facility designated by the procuring activity. Acceptance of the first article sample shall be based on no defects in the sample. Further production of the cartridge by the supplier prior to approval of the first article sample, shall be at the suppliers risk. Thread sealing compound may be omitted from the first article samples to facilitate disassembly for dimensional and hermetic testing of the cartridge.

* 4.3.2 Periodic production sampling. Periodic production tests detailed in 4.6 shall be performed at an activity designated by the procuring activity. Samples shall consist of 18 units from a maximum lot size of 1200 units that have satisfactorily passed the acceptance tests and examinations detailed in 4.7. Thread sealing compound may be omitted from the periodic production samples to facilitate disassembly for dimensional and hermetic testing of the cartridge.

* 4.3.2.1 Defective units. When a defective unit is found in any group, the supplier shall:

(a) Determine the cause of the defect.

(b) Determine if the defect is an isolated case.

(c) Take the necessary action to eliminate the defect from units on hand and, if so directed by the procuring agency, from those units previously shipped.

(d) Submit to the procuring activity for approval the proposed corrective action to prevent recurrence of the defect.

4.3.3 Acceptance sampling. Unless otherwise specified in the contract or order, a random sample of cartridges shall be sleected from each inspection lot for the testing required for lot acceptance (see 4.7.3). The size of this sample shall be not less than 50 cartridges, and shall conform to the requirements of MIL-STD-105 for an Acceptance Quality Level (AQL) of 0.25%.

4.4 Test conditions. Unless otherwise specified the cartridge shall be subjected to all tests specified herein under the following conditions:

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(a) Temperature — Room ambient (60 to 95°F)

(b) Altitude — Normal ground atmospheric pressure

(c) Humidity (relative) — Room ambient up to 95 percent

* 4.5 First article tests. The cartridges within the first article sample (identified as cartridge number 1 through 50) shall be subjected to the-tests specified in TABLE III. When specified in the contract (see 6.2, certified inspection data covering the results of first article tests shall be forwarded to the Naval Sea Systems Command. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests.

(b) Run or test plan used.

(c) Serial number of the cartridge

(d) Test results of first article tests.

4.5.1 First article test equipment. The following items of test equipment are required to perform the tests herein:

(a) Ohmmeter: 0 to 5 ohms, accuracy plus or minus 0.05 ohm at midrange, maxi.mum current 10 ma

(b) Megohmmeter: 500 vdc potential, current limited, commercial accuracy

(c) Oscilloscope: Dual beam, Tektronix Model 502 with C-12 Camera, or equivalent

(d) Power supply: 24-30 vdc (see FIGURE 2)

(e) power supply: 500 vdc

(f) Firing test circuit: (see FIGURE 3)

(g) Firing test setup: (see FIGURE 4)

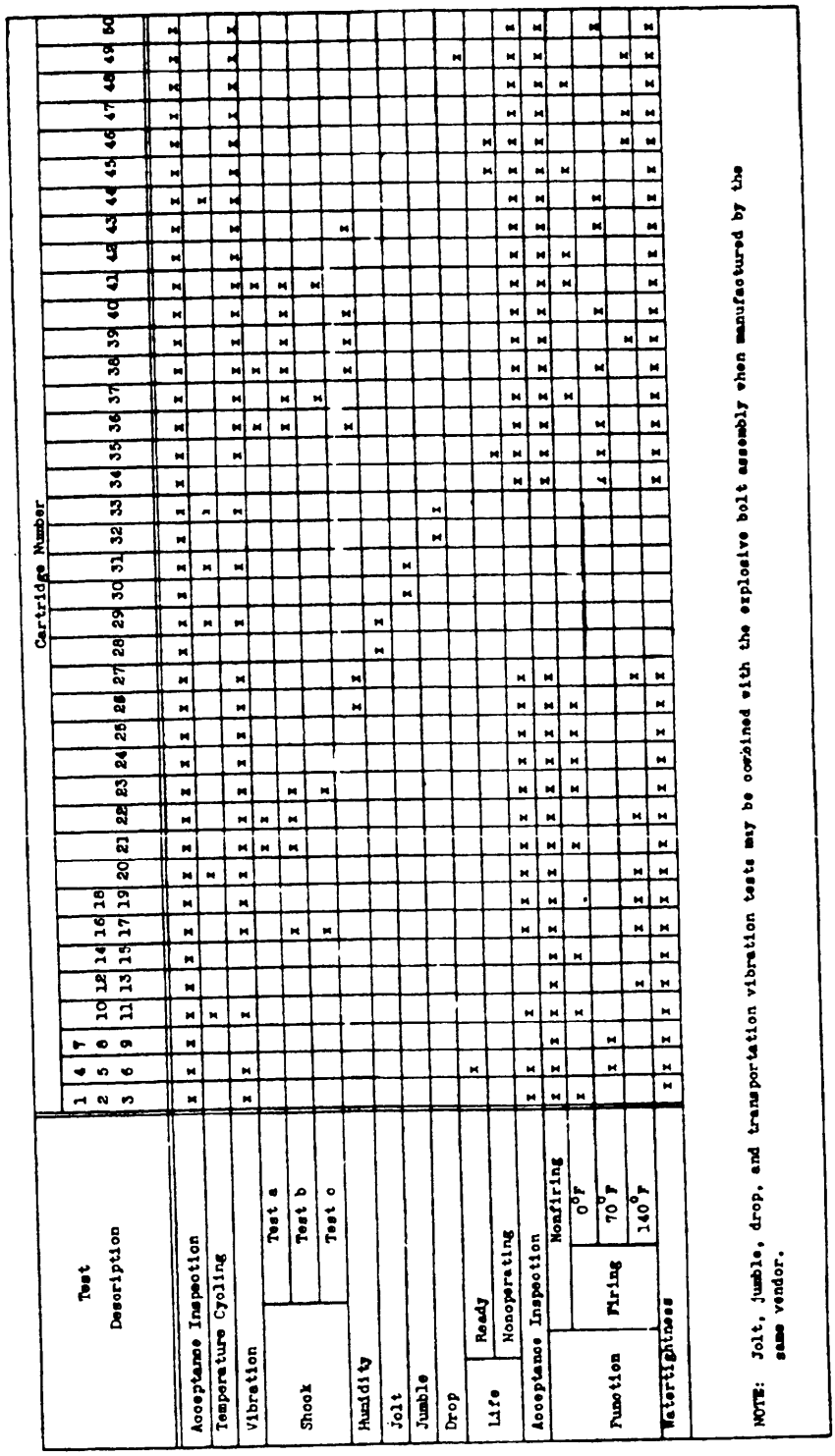
(h) Watertightness test setup: (see FIGURE 5)

4.5.2 Environmental tests. The cartridge shall be subjected to the following environmental tests (see TABLE 111). Unless otherwise specified, inability of the cartridge to pass the acceptance inspection and function tests, subsequent to the environmental tests, constitutes a failure.

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TABLE III

First article test breakdown



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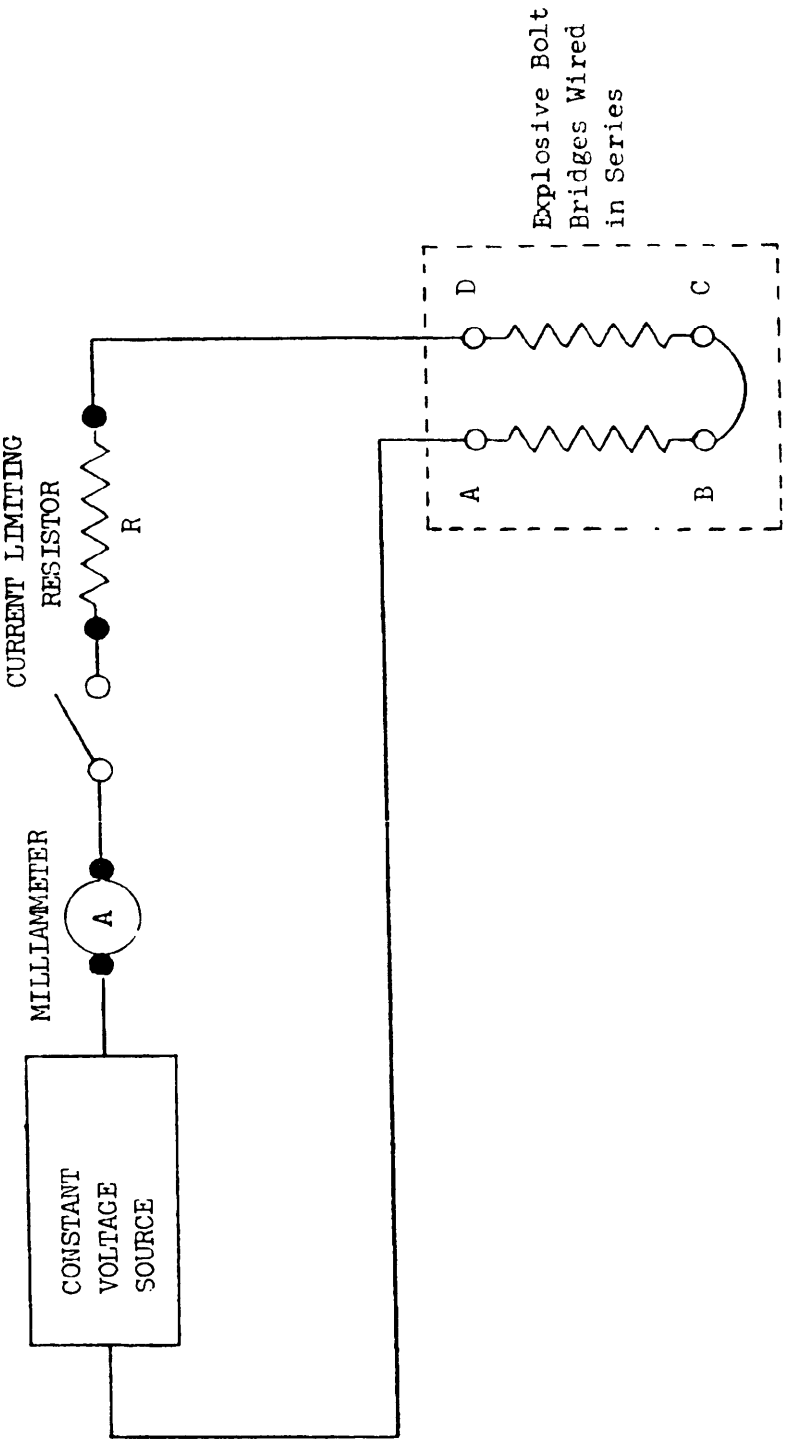


FIGURE 2. Nonfiring test circuit.

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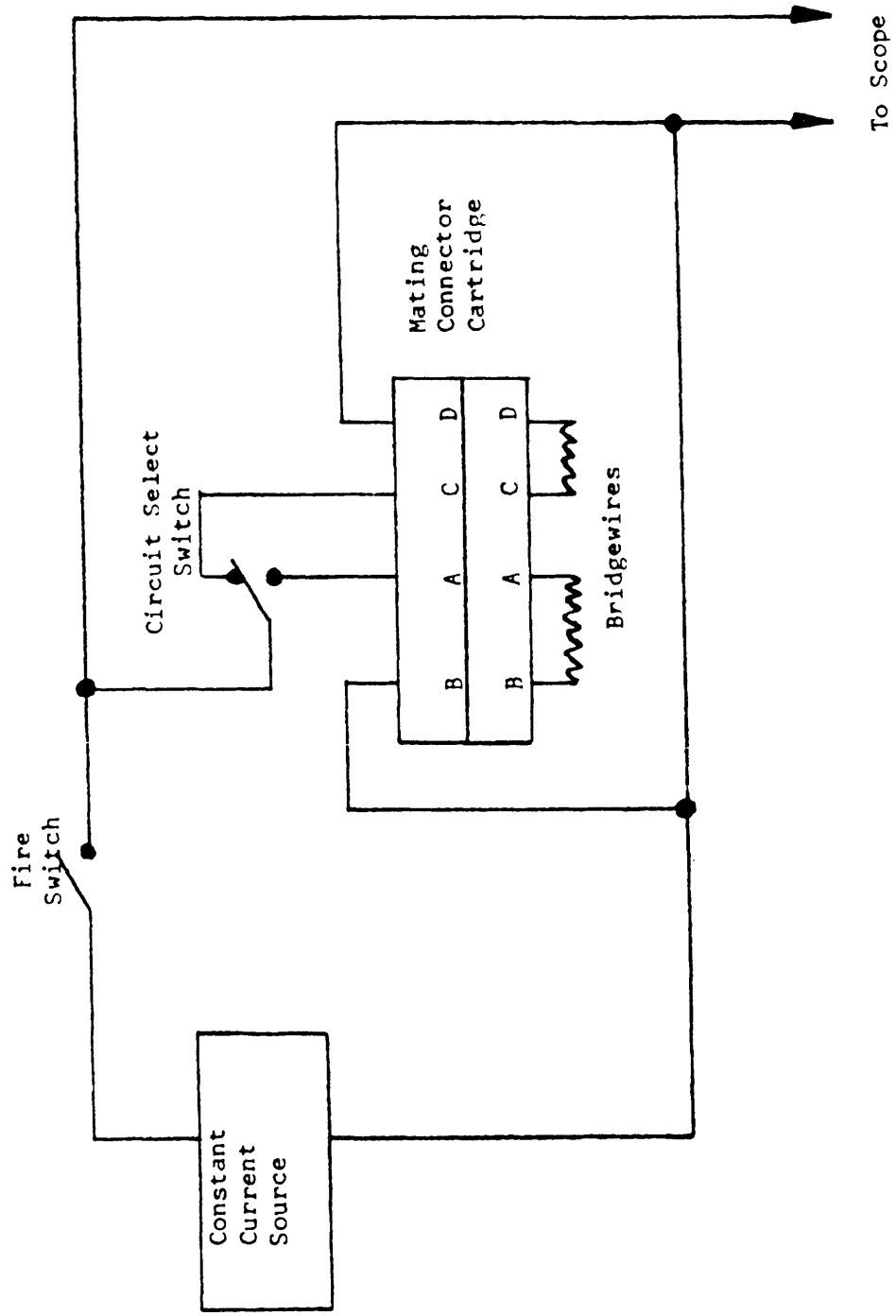


FIGURE 3. Firing test circuit.

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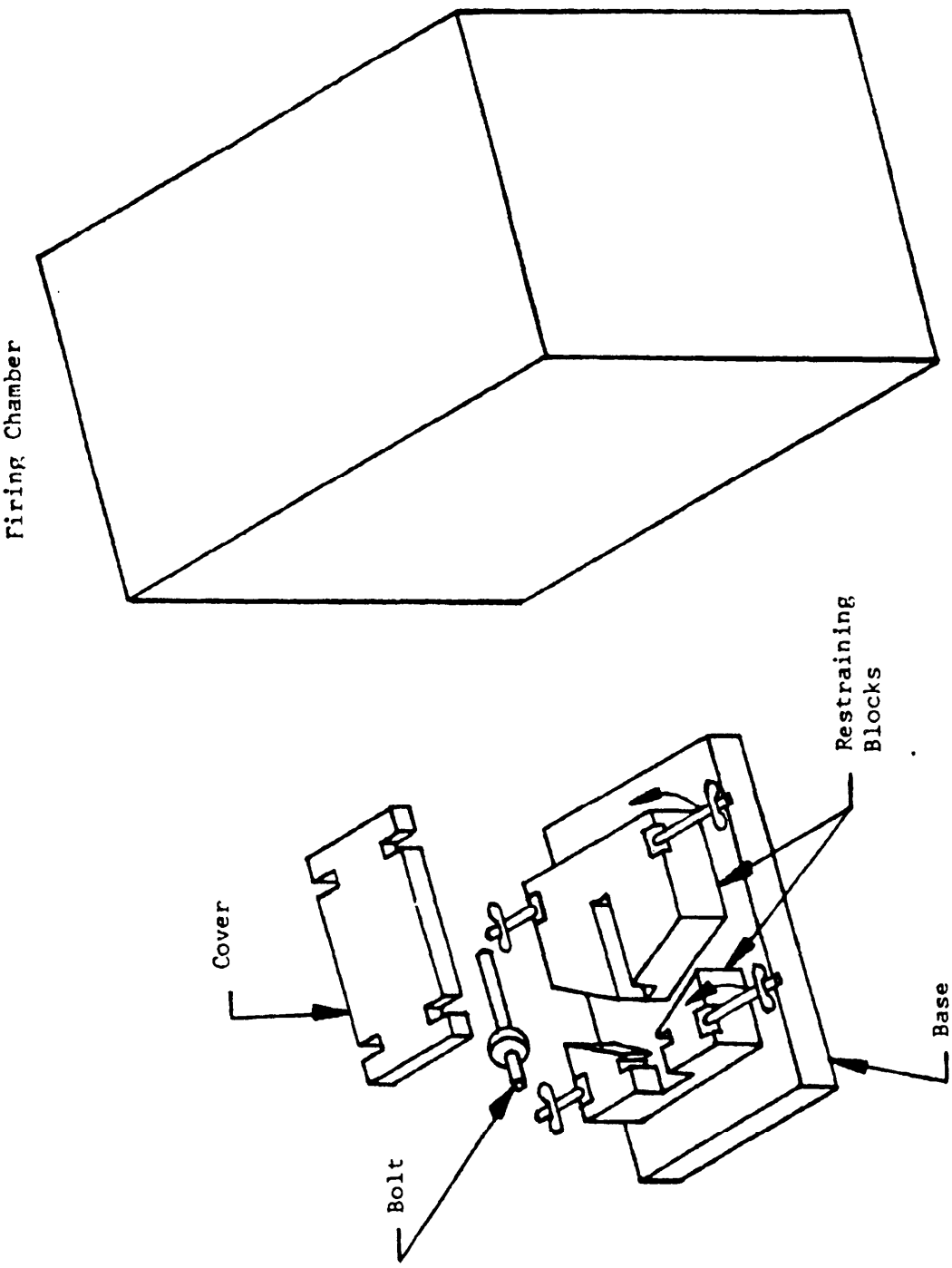


FIGURE 4. Firing Test Configuration.

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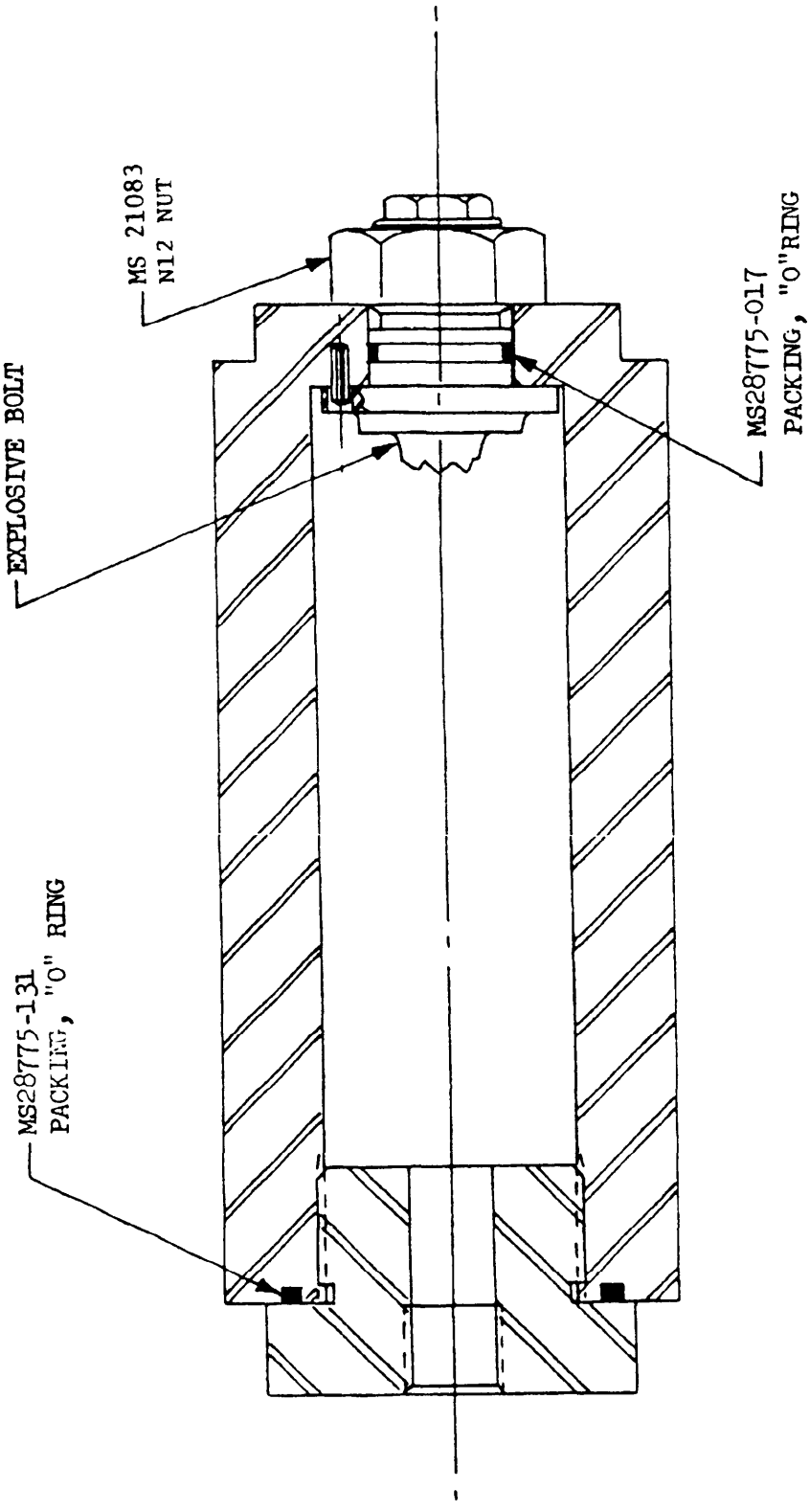


FIGURE 5. Watertightness Test Setup.

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4.5.2.1 Vibration. Mount the cartridge (see TABLE 111) into a fixture to simulate its service installation. The cartridge shall be conditioned at room temperature for a minimum period of one hour prior to conducting the resonance survey. Vibrate one third of the sample cartridges along each axis (see FIGURE 1) for a total of nine hours (3.4.1 and TABLE I) in accordance with the requirements of Method 514, MIL-STD-810.

4.5.2.2 Shock. Mount the sample cartridge, at room temperature, into a fixture to simulate its service installation. Apply prescribed shock so that a minimum of 90 percent of the table input to the fixture will be applied to the cartridge under test. Subject the cartridge to the shock listed in TABLE IV, as specified in Method 516, MIL-STD-810.

TABLE IV

Shock schedule

Test	Condition (see 6.3.5)	Peak magnitude (g's)	Duration (ins)	Impact direction (Axis-Fig 1)	Total number of impacts
a	Ready	300	0.5 to 1.5	Horizontal and vertical	6
b	Nonoperating	60	8.0 to 10.0	Any direction	9
c	Ready	45	20.0 to 21.0	Horizontal and vertical	6

4.5.2.3 Temperature cycling. The cartridge, preserved and packaged in accordance with 5.1.1, shall be placed in a temperature controlled chamber and subjected to the following temperature cycle:

(a) Cool to minus 45 plus 5 minus 0°F within 30 minutes

(b) Maintain at minus 45 plus 5 minus O°F for four hours

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(c) Heat to 140 plus 0 minus 5°F within 30 minutes

(d) Maintain at 140 plus 0 minus 5°F for four hours

(e) Return to room temperature within fifteen minutes. Repeat this schedule for a minimum of three cycles.

4.5.2.4 Humidity. The cartridge shall be tested in accordance with the humidity test of Method 507, MIL-STD-810 for eight hours at 110 plus or minus 5°F. The cartridge may be wiped dry before inspection.

4.5.3 Rough handling and safety. Cartridges from the first article sample (see TABLE III) shall be subjected to the following tests. Failure and rejection criteria shall be as specified in the applicable standards.

* 4.5.3.1 Jolt test. Cartridges from the first article sample shall be tested in accordance with MIL-STD-331, Test 101.

* 4.5.3.2 Jumble test. Cartridges from the first article sample shall be tested in accordance with MIL-STD-331, Test 102.

* 4.5.3.3 Forty-foot drop test. Cartridges from the first article sample shall be tested in accordance with MIL-STD-331, Test 103, except that one cartridge shall be dropped along the vertical axis and one cartridge shall be dropped along the longitudinal axis (see FIGURE 1), using Explosive Bolt 2132278 as the test vehicle.

4.5.4 Storage life, ready condition. The cartridge first article sample shall be subjected to 144 hours storage at 140 plus 0 minus 5°F. Inability of the sample, upon completion of storage, to pass the function test (4.5.6) ---- shall constitute a failure.

4.5.5 Storage life, nonoperating condition. The cartridge first article sample shall be subjected to 30 days storage at 140 plus 0 minus 5°F. Inability of the sample, upon completion of storage, to pass the function test (4.5.6) shall constitute a failure.

4.5.6 Function test. The function and watertightness tests shall be performed in a test chamber capable of maintaining the temperature within 5°F. The tests shall be performed in the following sequence to provide the maximum number of tests on each cartridge before destruction. During these tests, adequate safety provisions shall be enforced.

4.5.6.1 Nonfiring test. A current of 95 plus or minus 5 milliamperes shall be applied to the terminals of each circuit (see FIGURE 2) for 5 minutes minimum. Detonation of the cartridge shall indicate a defect.

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4.5.6.2 Firing test. Each cartridge to be tested shall be mounted in the fracturable bolt accepted to Dwg 2132278 and tested using the test circuit of FIGURE 3 and the test configuration of FIGURE 4. The firing signal shall be 0.5 plus or minus 0.02 ampere, applied alternately to the AB and CD bridgewire circuits in any group of functional test samples. The firing signal shall be monitored and the magnitude and duration of the signal recorded. The test samples shall be divided into three goups, in which one third of the samples shall be conditioned for four hours minimum at each of the following temperatures: O plus 5 minus 0°, 70 plus or minus 5°, and 140 plus 0 minus 5°F. Test samples shall be fired within two minutes after being removed from the conditioning chamber. Acceptance criteria for the firing test shall be as follows:

(a) A 15 millisecond miximum firing delay, as shown by the duration of the electrical signal.

(b) Detonation after ignition.

(c) Complete fracture of the bolt.

* 4.5.6.3 Watertightness. After firing, a hydrostatic pressure of 1000 psi shall be applied for one minute to the insert seal from the detonated end of the cartridge (see FIGURE 5). Any leakage past the insert seal may constitute a failure and cartridges exhibiting this leakage shall be additionally evaluated as follows:

A hydrostatic pressure of 1000 psi shall be applied for 10 minutes to the insert seal (see FIGURE 5). The cartridge in the fixture shall be inclined so that the connector end points downward. Any leakage past the insert seal that exceed one drop of water and which falls free shall constitute a failure.

* 4.6 Periodic production tests. The periodic production sample, selected in accordance with 4.3.2, shall be subjected to the tests of 4.6.2. When specified in the contract (see 6.2), certified inspection data covering the test results shall be forwarded to the Naval Sea Systems Command. Certified inspection data shall consist of the following:

(a) Items 4.5(a) through 4.5(c)

(b) Test results of periodic production tests

4.6.1 Periodic production test equipment. The following items of test equipment are required to perform the tests herein:

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(a) Power supply (see FIGURE 2)

(b) Power Supply (See 4.5.1(e))

(c) Firing test circuit (see FIGURE 3)

(d) Firing test setup (see FIGURE 4)

(e) Watertightness test setup (see FIGURE 5)

* 4.6.2 Periodic production test schedule. Subject the cartridges within the periodic production sample, selected in accordance with 4,3.2, to the tests and in the sequence indicated in TABLE V. The inability of the cartridge to pass any of the test criteria of TABLE V shall constitute a failure.

4.7 Acceptance tests and examinations. Acceptance tests and examination shall consist of two parts, as follows:

(a) Nondestructive tests and examination (see 4.7.2)

(b) Lot acceptance tests (4.7.3)

4.7.l Acceptance test equipment. The following items of test equipment are required to perform the acceptance tests herein:

(a) Ohmmeter (see 4.5.1(a))

(b) Megohmmeter (see 4.5.1(b))

(c) Power Supply (see FIGURE 2)

(d) Power supply (see 4.5.1(e))

(e) Firing test circuit (see FIGURE 3)

(f) Firing test setup (see FIGURE 4)

(g) Watertightness test setup (see FIGURE 5)

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TABLE V

Periodic test sequence

Test	MIL-C-83093 Para. No.	Sequence: No. samples:	A (3)	B (3)	C (3)	D (3)	E (3)	F (3)
Acceptance (We-firing)	4.7.2		1	1	1	1	1	1
Temperature (Cycling)	4.5.2.3						2	
Vibration	4.5.2.1						3*	
Shock	4.5.2.4						4*	
Humidity	4.5.2.4		2				5	
Life, Ready	4.5.4			2				
Life, Non-operating	4.5.5					2		
Non-firing	4.5.6.1		3	3		3	3	
Firing-one (1) each sample at 0°, 70° and 140°F	4.5.6.2		4	4		4	7	
Watertightness	4.5.6.3		5	5		5	8	
Jumble	4.5.3.2				2			
Drop	4.5.3.3							2*

*Visual checks after test

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4.7.2 Nondestructive tests and examination. Each cartridge to be delivered shall be subjected to the following tests and examinations. Failure to pass any test or examination specified herein shall be cause for rejection of the unit.

4.7.2.1 Circuit resistance. The cartridge shall be placed in a safety chamber and the resistance of each bridge circuit shall be obtained with a 5 milliampere current-limiting type ohmmeter. Any cartridge that exceeds the resistance limits of 3.3.5 shall be considered defective.

4.7.2.2 Insulation resistance. The insulation resistance of the cartridge shall be obtained using a megohmmeter with an integral source of 500 plus or minus 50 volts dc with an error of less than 10 percent. The resistance test shall be made using one pin of the cartridge and the metallic housing of the detonator cap as terminals. All four pins of the cartridge shall be shorted together with the cartridge contained in a suitable safety chamber during test. Any cartridge with an insulation resistance of less than two megohms shall constitute a failure.

* 4.7.2.3 Hermetic seal. Subject each cartridge to a leakage test in accordance with MIL-STD-202, Method 112B, Test Condition C (Tracer Gas), Procedure III(a). Do not, however, subject the cartridge to Test Conditions A or B (for Gross Leak Test). A leak rate in excess of 1×10^{-6} ATM cm^3/S shall be cause for rejection. The leak test shall be performed within 15 minutes maximum upon removal of the cartridge from the tracer gas environment.

4.7.2.4 X-ray examination. Each cartridge shall be identified and examined by radiographic techniques acceptable to the procuring activity. Loose powder, fissures or discontinuities between explosive materials, foreign or damaged material, and missing or dislocated parts shall be cause for rejection of the individual cartridges.

4.7.2.5 Examination of product. The cartridge shall be examined for conformance to the requirements specified herein and on Drawing 2158784. The presence of any one of the following defects shall constitute a failure:

(a) Threads not in conformance with FED-STD-H28 and drawing requirements

(b) Electrical igniter unit loose

(c) Identification and marking not located as specified, not legible, or data incomplete

(d) Presentation, packaging, or packing not in accordance with Section 5

(e) Terminal pins not shorted

(f) Visible holes, perforations, or cracks in 0.315" diameter casing of detonator cap

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4.7.2.5 Examination of product (Continued)

(g) Evidence of corrosion

(h) Evidence of pyrotechnic material on threads and casing of cartridge

(i) Cartridge not complete; component parts missing

(j) Component parts not new; evidence of reclaimed, used, or salvaged components

(k) Cartridge configuration extends beyond drawing limits; dimensions do not comply with drawing requirements

(l) Surface finsihes or coatings damaged or incomplete

(m) Evidence of unauthorized material and repairs

4.7.3 Lot acceptance tests. Unless otherwise specified, the cartridges within the acceptance cample (see 4.3.3) shall be sujected to the following tests. Lot acceptance shall be based on no defect in the sample.

4.7.3.1 Moisture content of intermediate and base charges. An approximate two grain sample of the intermediate and base charge material shall be collected at the-site of, and immediately prior to, loading the material into the cartridge. The sample shall be palced in a tared, stoppered container, weighed and then opened. The sample shall then be exposed to a temperature of 126 plus or minus 4°F for drying and a minimum vacuum of 28 inches mercury until a constant weight is reached. The container shall then be cooled in a desiccator, stoppered and weighted. All weighings shall be accurate to plus or minus 0.005 milligram. The presence of moisture in excess of 1.0 percent in the intermediate charge and 0.13 percent in the base charge shall constitute a failure.

4.7.3.2 Function. Divide the acceptance sample into two groups. The first group shall consist of three cartridges. The second group shall consist of the balance of the acceptance sample. Subject each cartridge within the second group to the function test of 4.5.6. Inability of the cartridge to meet the function test of 4.5.6 shall constitute a failure.

* 4.7.3.3 Rough handling and safety tests. Subject the cartridges within the first group to the following tests. Failure criteria shall be as specified in the applicable standard.

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(a) Jolt Test: One third of the cartridges within the first group shall be jolt tested in accordance with MIL-STD-331

(b) Jumble test: One third of the cartridges within the first group shall be jumble tested in accordance with MIL-STD-331

(c) Drop test: One third of the cartridges within the first group shall be subjected to the 40-foot drop test of 4.5.3.3

5. PACKAGING

5.1 Preservation and packaging. The unit package shall contain one cartridge.

* 5.1.1 Level A. The cartridge shall be preserved and packaged in accordance with Code of Federal Regulations 49 CRF 171-178.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Not applicable.

5.2 Packing

* 5.2.1 Level A. The cartridge, preserved and packaged in accordance with 5.1.1, shall be packed in a container in accordance with Code of Federal Regulations 49 CFR 171-178.

5.2.2 Level B. Not applicable.

5.2.3 Level C. Not applicable.

* 5.3 Marking. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 and Code of Federal Regulations 49 CFR 171-178.

6. NOTES

6.1 Intended use. The cartridge is intended for use in Mk 46 torpedoes. Detonation of the cartridge seines to release lead ballast from an exercise torpedo by fracturing the ballast retaining bolt.

* 6.2 Ordering data.

6.2.1 Acquisition requirements. Procurement documents should specify the following:

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(a) Title, number, and date of this specification

(b) Type cartridge (see 1.2)

(c) If a first article is not required (see 3.1)

(d) Assigned activity for first article and periodic tests (see 4.3)

(e) If certified inspection data are required (see 4.5, 4.6)

(f) Any exceptions to this specification

(g) That all additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any cartridge(s) to pass the prescribed periodic production inspections.

(h) That the safety precaution requirements of the contractor's Safety Manual for Ammunition, 'Explosives and Related Dangerous Material, DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by the Government Activity, the safety precaution requirements of Ammunition Ashore, OP5, are applicable.

* 6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data item description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data item specified below shall be delivered by the contractor in accord- and with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirement title	Applicable DID No.	Option
4.5 and 4.6	Inspection and Test Reports	DI-T-5329	------

(Data item descriptions related to this specification and identified in Section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of 50 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

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6.3 Definitions

6.3.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second(hertz).

6.3.2 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's), duration in units of time, milliseconds (ins), and a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines, tangent at maximum slope, intersect the zero g line.

6.3.3 Pressure. Unless otherwise specified, pressure is the pressure of the environment to which the cartridge is exposed, expressed in pounds per square inch absolute (psia) or pounds per square inch gage (psig).

6.3.4 Cartridge conditions

6.3.4.1 Nonoperating. The condition of the uninstalled cartridge when stored, transported, or handled.

6.3.4.2 Ready nonoperating. The condition of the cartridge when assembled with the end item, transported, and handled.

6.3.4.3 Ready. The condition of the cartridge when fully assembled with the torpedo and installed on the launching mechanism prior to operating.

6.4 Preparation for delivery criteria. Preservation, packaging, and packing shall be Level A only. This level is used for those items which are to shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

* The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the require ments of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-81093B(OS)	CARTRIDGES, EXPLOSIVE BOLT (For Torpedoes MK 46 All Mods)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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MILITARY SPECIFICATION

CARTRIDGES, EXPLOSIVE BOLT

(For Torpedoes Mark 46 all Mods)

This amendment forms a part of Military Specification MIL-C-81093B(0S) dated 13 September 1984, and is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2. Delete: Applicable Documents substitute "Government documents".

2.1 Delete: "Government Documents and substitute:

"2.1 Government documents.

"2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

2.1.2 Delete entire paragraph and substitute the following:

"2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation."

At the bottom of page, delete and substitute:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640-5000

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3.3.3.2 Delete and substitute:

* "3.3.3.2 Intermediate charges. The second initiating agent shall consist of polyvinyl alcohol precipitated lead azide (RD 1333 or PVA lead azide) having a moisture content of not more than 1.0 percent."

Preparing Activity:
Navy - OS
Project No. 1356-N027

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